Energy Reports 9 (2023) 3380–3396

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Techno-economic Assessment of a Hydrogen-based Islanded Microgrid in North-east

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ARTICLE INFO

Article history:
Received 21 October 2022
Received in revised form 7 February 2023
Accepted 8 February 2023
Available online 21 February 2023

Keywords:
Electrolyser
Fuel cell
Hydrogen energy
Islanded power system
Microgrid
Solar PV
Wind

ABSTRACT

Currently, renewable energy-based generators are considered worldwide to achieve net zero targets. However, the stochastic nature of renewable energy systems leads to regulation and control challenges for power system operators, especially in remote and regional grids with smaller footprints. A hybrid system (i.e., solar, wind, biomass, energy storage) could minimise this issue. Nevertheless, the hybrid system is not possible to develop in many islands due to the limited land area, geographical conditions, and others. Hydrogen as a carrier of clean energy can be used in locations where the installation of extensive or medium-scale renewable energy facilities is not permissible due to population density, geographical constraints, government policies, and regulatory issues. This paper presents a techno-economic assessment of designing a green hydrogen-based microgrid for a remote island in North-east Australia. This research work determines the optimal sizing of microgrid components using green hydrogen technology. Due to the abovementioned constraints, the green hydrogen production system and the microgrid proposed in this paper are located on two separate islands. The paper demonstrates three cost-effective scenarios for green hydrogen production, transportation, and electricity generation. This work has been done using Hybrid Optimisation Model for Multiple Energy Resources or HOMER Pro simulation platform. Simulation results show that the Levelized Cost of Energy using hydrogen technology can vary from AU$0.37/kWh to AU$1.08/kWh depending on the scenarios and the variation of key parameters. This offers the potential to provide lower-cost electricity to the remote community. Furthermore, the CO2 emission could be reduced by 17,607,77 kg/year if the renewable energy system meets 100% of the electricity demand. Additionally, the sensitivity analysis in this paper shows that the size of solar PV and wind used for green hydrogen production can further be reduced by 50%. The sensitivity analysis shows that the system could experience AU$0.03/kWh lower levelized cost if the undersea cable is used to transfer the generated electricity between islands instead of hydrogen transportation. However, it would require environmental approval and policy changes as the islands are located in the Great Barrier Reef.

1. Introduction

There is a worldwide trend towards increasing the share of renewable energy in power and energy system. Significant work has been done for the cost-effective and optimal placement of renewable-based distributed generation and energy storage into the system (Yang et al., 2021, 2020). Hydrogen has become popular in the renewable energy sector due to its potential for storage and transportation, similar to conventional fuels. The stored hydrogen can then be used in fuel cells and/or combustion engines to generate electricity (Felseghi et al., 2019). One kilogramme of hydrogen contents of 33.36 kWh of useable energy. It is one of the materials with the highest gravimetric energy density (Handwerker et al., 2021). A study on hydrogen fuel cells by (Manoharan et al., 2019) reported that one kilogramme of hydrogen has a higher density than most batteries. Almost 90% of the global hydrogen is currently produced from SMR (He and Li, 2014). This is not an environmentally friendly process of producing hydrogen due to the high CO2 emissions. However, SMR is currently

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https://doi.org/10.1016/j.egyr.2023.02.019

Abbreviation

Indices

AC	Alternating Current
ARENA	Australian Renewable Energy Agency
AU$ or AUD	Australian Dollar
BOM	Bureau of Meteorology (Australia)
CAPEX	Capital Expenditure
CO2	Carbon Dioxide
DC	Direct Current
DG	Diesel Generator
EV	Electric Vehicle
H2	Hydrogen
HOMER	Hybrid Optimisation of Multiple Energy Resources
IRENA	International Renewable Energy Agency
kg	Kilogramme
kW	Kilowatt
kWh	Kilowatt-hour
LCC	Life Cycle Cost
LCOE	Levelized Cost of Energy
LCOH	Levelized Cost of Hydrogen
MW	Megawatt
NASA	National Aeronautics and Space Administration
NPC	Net Present Cost
NPCh	Net Present Cost associated with hydrogen production
NPV	Net Present Value
O&M	Operation and Maintenance
PV	Photovoltaic
R&D	Research and Development
RE	Renewable Energy
RES	Renewable Energy Source
SMR	Steam Reforming Process
US$ or USD	US Dollar

the cheapest hydrogen production method (Yilmaz et al., 2015). Hydrogen can also be produced by the electrolysis of water using renewable energy sources (RES). A study by Timmerberg et al. (2020) reported that the CO2 emissions from hydrogen production could be reduced significantly with RES powered electrolysis process (known as green hydrogen production). The on-site green hydrogen production would reduce more emissions (up to 99%) than other methods, such as hydrogen gas blending with other gases or mixing with a liquid hydrogen supply chain (Gu et al., 2020). The economic potential of wind-based hydrogen production is discussed by Genç et al. (2012). The future perspectives of solar-based hydrogen production are outlined in Ozalp et al. (2010). Hydrogen can also be produced using other alternative sources. The production of hydrogen from biomass has been reported by Hosseini et al. (2015). Furthermore, the production of hydrogen generation cells for microalgae has been discussed in Göllei et al. (2016) and Gholkar et al. (2021), respectively. Green hydrogen has the potential to overcome constraints associated with RES, such as intermittency and long-term energy storage issues (Kovač et al., 2021). Moreover, hydrogen can also be transported to locations where RES installation is impossible. Therefore, hydrogen consumption is anticipated to increase in the near future. Global market growth of hydrogen is expected to reach 120 million tonnes by 2024 (60 million tonnes more than the 2019 level) (Atilhan et al., 2021). By 2070, it is predicted that hydrogen demand will increase significantly and take over 50% of the market share (Moreno-Benito et al., 2017).

A feasibility study of a hybrid system for electricity generation and green hydrogen production in the Hendijan area of Southwest Iran has been reported (Qolipour et al., 2017). However, this research work has mainly focused on a grid-connected system with solar and wind generation systems. A feasibility study of a green hydrogen system in an islanded microgrid was reported for the Canadian Arctic Community of Sanikiluaq (Vera et al., 2020). This article uses a HOMER model to represent the system. A lower hydrogen-based electricity cost in comparison to the base system was reported in this study. However, the hydrogen system is considered as storage in this work. Another research work proposed by Temiz and Dincer (2021), has reported a green hydrogen production system using HOMER for short-distance ferries in Lake Ontario, Canada. This study reported the production of 26.7 tonnes of green hydrogen from grid-connected floating solar panels (Temiz and Dincer, 2021). However, this work mainly focused on generating hydrogen fuel for marine vessels. In Singh et al. (2017), the economic feasibility of a standalone hybrid system has been reported. This study was conducted in Bhopal, India. This has considered the green hydrogen system and solar PV systems to meet the electricity demand of a research facility. The LCOE of the hydrogen-based off-grid system is reported in Colorado, USA (Abdin and Mérida, 2019). The study suggests the long-term techno-economic benefits of hydrogen over batteries for an isolated power system. A solar-based hydrogen production map for Turkey has been reported by Karayel et al. (2022a). The analysis estimated that Turkey could potentially produce 415 to 427 million tonnes of hydrogen from solar energy per year. The study also showcased the perspectives of using different types of electrolysers. Another study on the total hydrogen production potential of Turkey with wind resources was outlined by Karayel et al. (2022b). The country can potentially produce 251 million tonnes of hydrogen per year as an alternative energy storage option after meeting the demand of all major cities (Karayel et al., 2022b). However, these studies focused mostly on determining the potential sources for hydrogen generation without considering underlined constraints.

A floating solar PV-based electricity network with a green hydrogen storage system was designed and analysed for Mugla Province, Aegean Region of Turkey (Temiz and Javani, 2020). The NPC and the LCOE were considered for the optimisation process. The hybrid system has been reported to meet 99.43% of electrical demand without grid connection and fossil fuel-based generation with an LCOE of US$ 0.612/kWh (Temiz and Javani, 2020). However, the focus was mainly on hydrogen-based storage for the system. Green hydrogen has also been produced and utilised in the transportation sector, where fuel cells were used to energise the drivetrains of EVs (Silvestri et al., 2021). The LCOH was found to be € 9.02/kg, i.e., approximately AU$ 14.3 per kg. Furthermore, hydrogen production from wind and PV for a self-sustain fuelling station was simulated (Zhao and Brouwer, 2015). The study found a per kg hydrogen production cost of AU$ 9.71 and AU$ 13.23 for a wind and solar-based system, respectively. Another study evaluated the LCOH for a decentralised hydrogen refuelling station reported in Belgium. The LCOH is reported to be AU$ 15.26/kg for such a refuelling station (Viktorsson et al., 2017). Furthermore, a comparative analysis of producing PV-powered green hydrogen reported an LCOE range between AU$ 9.1/kWh and AU$ 17.93/KWh (Shaner et al., 2016). According to the literature, solar and wind-powered hydrogen production costs vary between AU$ 9 and AU$ 18 per kg. A hybrid electric sharing system in California is reported by He et al. (2021b,a).

Fig. 1. Geographical location of Torres Strait.

The synergistic interaction of thermal, electrical and hydrogen systems has been considered in a community microgrid. An EV interaction with the grid was also considered. Further, energy sharing between the microgrid and hydrogen vehicles was reported by He et al. (2021b,a). This work considered the residential community for the energy-sharing study. However, most of these studies have focused on hydrogen production with no indication regarding hydrogen transportation or the use of hydrogen in fuel cells for electricity generation.

Australia has geographical advantages to produce green hydrogen. The Renewable Hydrogen Development (R&D) Funding Rworth of AU$70 million was announced by ARENA in 2019 to fast track the green hydrogen development (ARENA, 2022; Shillington and Brady, 2022). The Hydrogen Funding Round (R&D) has already provided AU$ 22.1 million funding by the 2022 (ARENA, 2022). Additionally, a number of funding initiatives were introduced in different Australian states. The state government of Queensland has announced a Hydrogen Industry Development Fund (HIDF) of AU$ 25 million over a four-year period (Shillington and Brady, 2022). This fund will be allocated to eligible projects in two streams. These are: the plant and equipment stream (Funding Stream One) to provide financial assistance of maximum 50% of project investment, whereas the feasibility stream (Funding Stream Two) would assist with maximum of AU$ 500,000 (up to 25%) for feasibility study (Shillington and Brady, 2022). The HyP SA in South Australia is the first project in Australia to blend the green hydrogen with gas network. This project is managed by the Australian Gas Infrastructure Group with financial support from South Australian government. The project blends green hydrogen with natural gas (up to 5% hydrogen volumes). The natural gas blended with 5% hydrogen is then supplied to 700 homes in South Australia. This project has a 1.25 MW electrolyser to produce 20 kg hydrogen per hour (Australian Gas Infrastructure Group, 2022). Australian Gas Infrastructure Group is also considering to expand green hydrogen projects in Gladstone, Queensland. This project would consider 10% hydrogen to be blended with natural gas. Therefore, more green hydrogen-based projects are expected, not only in Queensland but also in other parts of Australia.

The Thursday Island is one of the major islands of the Torres Strait in between Australia and Papua New Guinea (PNG). The Cape York Peninsula is located to the south of Torres Strait, the Northernmost location of the Australian mainland. Fig. 1 shows the geographical location of Torres Strait. Thursday Island is one of the few islands in the Torres Strait with inhabitants. The area of Thursday Island is only 3.5 km^2 and its population is approximately 3000 (Britannica, 2022; Queensland Government, 2018). It is the most densely populated island in Torres Strait. The island acts as the administrative and commercial hub of Torres Strait (Queensland Government, 2018). Ergon Energy Network, a distribution network service provider in Queensland, operates and maintains electricity generation and distribution in Torres Strait.

This study aims to reduce the dependency on diesel for electricity generation by integrating RES and energy storage systems into Thursday Island. Furthermore, it explores the potential impacts of RES integration on the costs of energy production on the island. However, the limited land area is the primary constraint to integrating RE facilities into the island. To overcome this obstacle, the production of green hydrogen on the neighbouring island (Horn Island) could be considered. Horn Island has located approximately 2.5 km from Thursday Island, with a potential area of 53 km ^{2} for green hydrogen production. The population on this island is small (approximately 550 people) (Centre for The Government of Queenland, 2022). The island is mainly used as a gateway to the Torres Strait due to the airport facility. Fig. 2 visually compares these two islands in terms of area. The focus of this research project is to conduct a feasibility study for green hydrogen on Horn Island by electrolysis process and transport the hydrogen to Thursday Island for electricity generation using hydrogen fuel cells. The objectives of this study are given next:

- To design a least-cost model for the green hydrogen-based microgrid.
- Propose a feasible infrastructure concept design based on cost optimisation.

Producing hydrogen in the form of gas is cheaper than liquid hydrogen, whereas the transportation of liquid hydrogen is comparatively cheaper (Almansoori and Shah, 2009, 2012). However, the modes of transportation can be selected based on the types of hydrogen. For example, railway tanker cars, tanker trucks and ships are the better options for transporting liquid hydrogen, whereas gaseous hydrogen can be transported via pipelines, tube trailers, and railway tube cars (Güler et al., 2021). Transporting hydrogen through pipelines generates excellent potential in the hydrogen economy, as reported by Cerniauskas et al. (2020). Hydrogen gas pipeline with gaseous trailer distribution is a new area of research and development. The hydrogen transmission pipelines will be a key to a cost-effective green hydrogen supply chain in future (Reuß et al., 2019). Moreover, development phases require several transitions, e.g., production scaling and a combination of different transportation modes (Moreno-Benito et al., 2017). Existing natural gas pipelines can also be used for transportation. This is considered a cheaper option, as reported by Cerniauskas et al. (2020). However, only 10% of the capacity

Fig. 2. Geographical area comparison of Thursday Island and Horn Island.

Table 1
Summary of literature review.

Reference	Microgrid	Hydrogen as the primary source	Hydrogen transportation	Unmet hydrogen	Other options such as HVDC	Remote community
Qolipour et al. (2017)	×	×	×	×	×	×
Vera et al. (2020)	√	×	×	×	×	×
Singh et al. (2017)	√	×	×	×	×	×
Temiz and Javani (2020)	√	×	×	×	×	×
Abdin and Mérida (2019)	√	√	×	×	×	×
He et al. (2021b,a)	√	√	–	×	×	×
Proposed	√	√	√	√	√	√

of the gas pipelines can be used in such cases (Clees et al., 2021). Since there is no existing pipeline between Thursday and Horn islands, the cost of hydrogen transportation must be incorporated for optimisation.

The comparison of the state-of-art literature is given in Table 1. It is noticed in Table 1 that no work in the literature has considered the production and use of hydrogen in two separate islands for microgrid systems. Moreover, most of the works have considered hydrogen as a backup or alternative energy storage. Comparative comparison of various options and sensitivity of key parameters on LCOE and LCOH was missing in many works.

A number of techno-economic studies have been reported in the literature for green hydrogen production. However, this is a novel work conducted in the Australian Torres Strait due to the geographical uniqueness of the location. Moreover, to the best knowledge of the authors, this is the first study of a green hydrogen-based islanded microgrid where electrolysers and fuel cells are situated at two different locations. The findings of this study will also contribute as a guideline to place hydrogen-based microgrids not only in Australian isolated islands but also in similar global remote locations e.g., small isolated island groups in Indonesia, Philippines, and Pacific Ocean countries where a number of remote energy systems are situated in close proximity. The main contributions of this work are outlined below:

- A proper techno-economic model of a real islanded microgrid based on actual data, wind, and solar profiles has been developed. In this model, green hydrogen is considered the primary fuel source to power the islanded microgrid.
- This study has considered the placement of electrolyser and fuel cells in separate islands.
- Optimal solar and wind generator sizing has been done using the different levels of unmet hydrogen.
- Potential investment in hydrogen production is determined considering the uncertainties in electrolyser characteristics, PV system performance, and unmet hydrogen for various realistic scenarios.
- The impact of hydrogen transportation(through short-distance pipelines) on LCOE of the system has been assessed. Also, this work compared the techno-economic performance of a fuel cell-based system on the island against the system electrically connected via sub-sea cables.

The rest of the paper is organised as follows: Section 2 describes the overview of the current system and an illustration of the overall work. The methodology of this work is explained in Section 3. The results, discussions, and comparisons are given in Section 4. Section 5 outlines the conclusion of this work.

2. Overview of the system

The electrical network of Thursday Island is an islanded microgrid. The daily average electricity energy demand on the island is 67,692 kWh. The average hourly power demand is 2820.5 kW. The hourly peak demand could reach up to 4062 kW in the summer. Fig. 3 presents the geographical location of the existing system on Thursday Island, whereas Fig. 4 illustrates the daily load patterns of Thursday Island. The similar time series data in Fig. 4 are used for the optimal system design.

The existing electricity generation on Thursday Island mainly relies on diesel generators. Currently, there are four diesel generators with a total capacity of 9.1 MW. There are also two wind

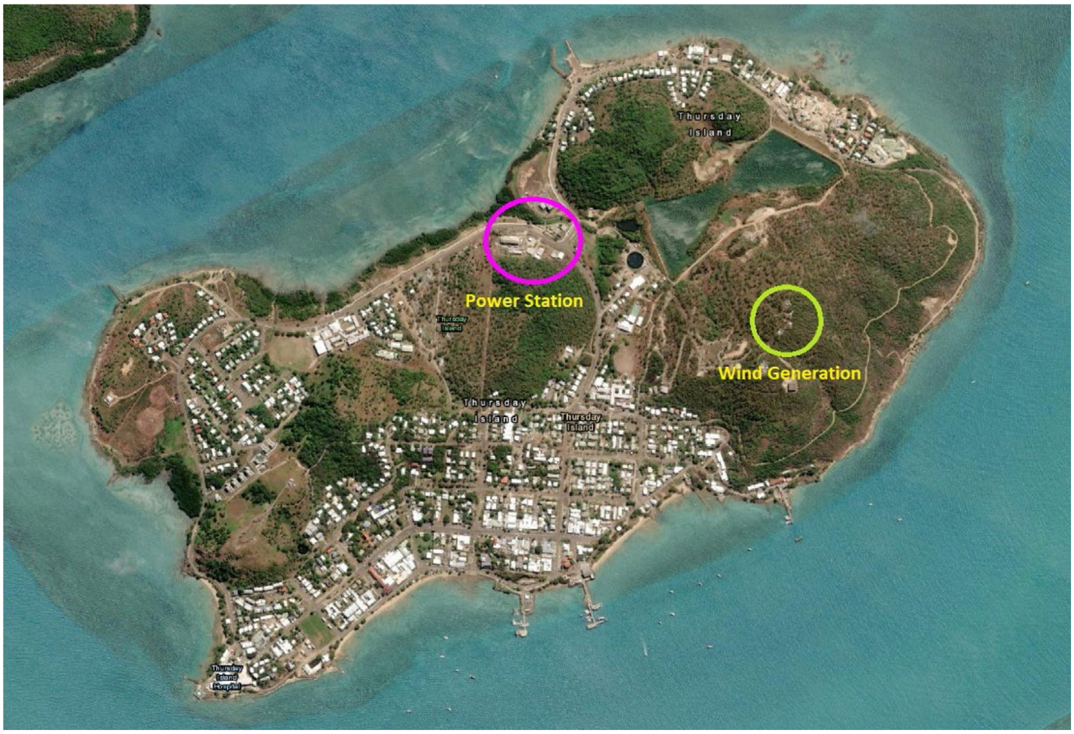

Fig. 3. Geographical location of the generation systems within Thursday Island.

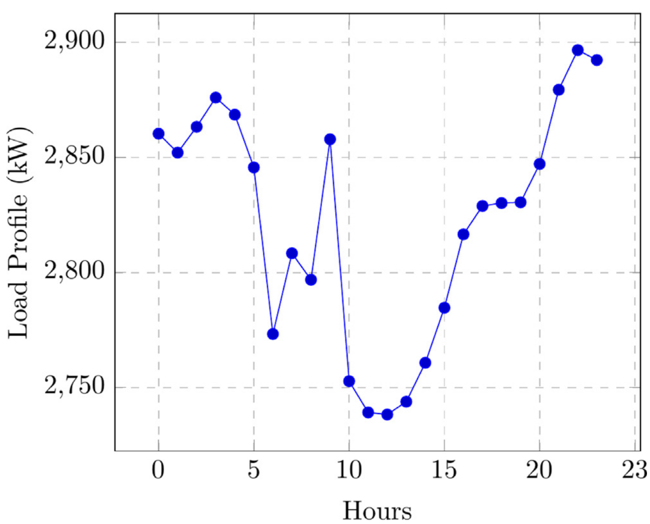

Fig. 4. Average daily electrical load of Thursday Island.

generators with a total capacity of 450 kW. Accordingly, Thursday Island's instantaneous renewable energy capacity ranges between 5% to 10% of total generation. Table 2 presents the existing generation components of Thursday Island. The aggregated load and diesel/wind generation units on Thursday Island are illustrated in Fig. 5.

The current system on Thursday Island has higher costs per kWh than on the main eastern seaboard grid due to remoteness, smaller generation scale and diesel fuel costs. The reliance on diesel generators must be reduced to meet the net zero target set by Queensland state and the Australian federal government. However, geographical location and availability of land are the main constraints for integrating more renewables into this system.

The area of Thursday Island is too small for centralised renewable energy infrastructure apart from rooftop solar PV in homes and businesses. To meet the entire electricity demand using RES, majority of the land on this island must be used. Therefore, the only feasible option is transferring energy from other locations (currently in the form of diesel).

Table 2
Existing generation units.

Existing generation units	Capacity
Diesel Genset (DG1)	2.7 MW
Diesel Genset (DG2)	2.7 MW
Diesel Genset (DG3)	2.7 MW
Diesel Genset (DG4)	1.0 MW
Wind Turbine (W1 & W2)	450 kW

Based on the challenges mentioned earlier, the first approach is to discern potential clean energy sources nearby. Horn Island is a larger island with a small population. Therefore, it can be used for green hydrogen production. Furthermore, the hydrogen could be transported to Thursday Island to be utilised in fuel cells to generate electricity. This study minimises the cost of producing the required hydrogen on Horn Island and transfers the green hydrogen for electricity generation using fuel cells on the Thursday Island microgrid. Three scenarios have been proposed in this article to perform the analysis. Table 3 illustrates the three various scenarios used in this study. A combination of different penetration levels of renewable and diesel generations have been considered under these three scenarios. In all of the scenarios, it is assumed that 10% of the electricity can be generated from the existing wind turbines on Thursday Island. For example, the 100% renewable energy scenario shown in Table 3 means 90% of the required electricity can be generated from hydrogen fuel cells, and the rest of the demand(i.e., 10%) could be met by the existing wind generators. Therefore, this study aims to estimate the amount of required green hydrogen on Thursday Island for electricity generation as well as the optimal sizing of solar PV and wind generators on Horn Island required to produce hydrogen.

Due to the geographical constraints of this project, the energy needs to be transferred between the two islands. A study in

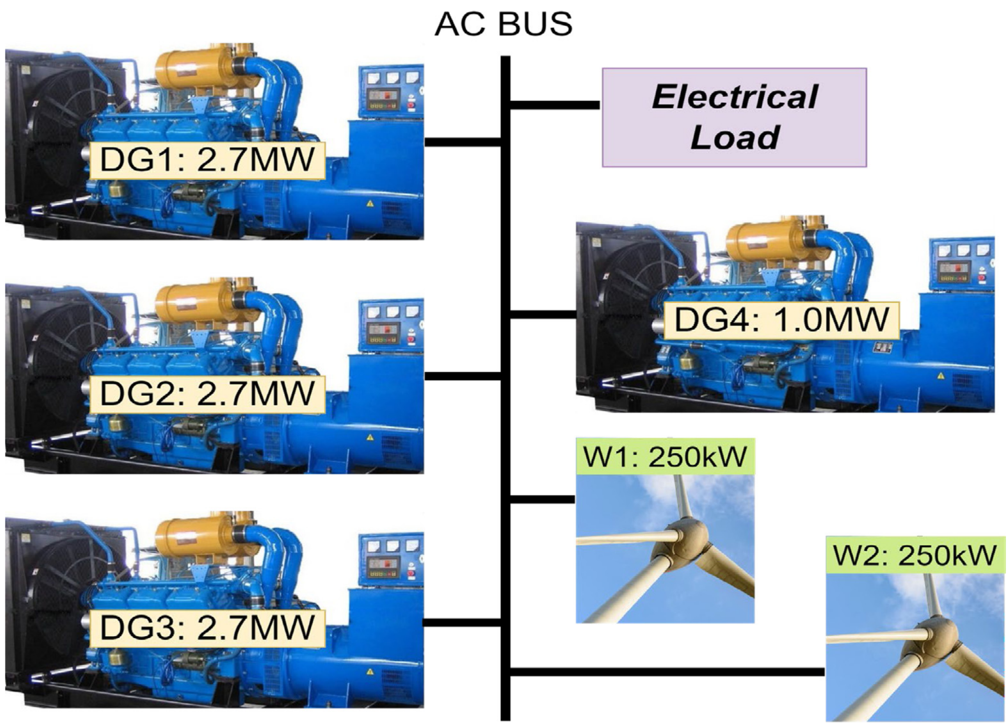

Fig. 5. Existing electricity generation system on Thursday Island.

Table 3
Three scenarios used in this project.

Renewable energy penetration level	Existing diesel generator
Up to 100% (wind 10%)	Backup only
80% (wind 10%)	20%
50% (wind 10%)	50%

Bertheau and Cader (2019), reports that undersea cables connecting 35 islands in the Philippines can reduce generation costs. The study also reported that undersea cable connection is a cost-effective solution, although the cabling cost increases by 90% to connect short distant islands. Another study on Dogger Bank Round 3 offshore wind project in UK concludes that high-voltage DC is more cost-effective for long distant undersea cable connections (Nieradzinska et al., 2016). It is reported in Nieradzinska et al. (2016) that the cost of installing undersea cables can be AU$1.75 million per km. However, installing undersea cables in a remote location like Torres Strait can be challenging. The Great Barrier Reef and reef-related marine life surround Torres Strait islands. Due to this constraint, using undersea cables to transmit renewable-based electricity between Horn and Thursday islands is prohibited. Moreover, there are currently no existing undersea cables or gas pipelines between Torres Strait islands. Furthermore, this research work contributes to Ergon Energy's Hydrogen Hubs powering remote communities (H2H) project. The underlying instruction for this project is to consider a hydrogen supply chain option rather than powering Torres Strait islands through undersea cables. Because it is potentially possible in future to supply hydrogen as a clean fuel to the other Torres Strait islands as well from this facility, those islands are located remotely with fewer populations where installing undersea cable will not provide a return on the investments.

Based on a report published on May 21 2021, by Clean Energy Finance Corporation and Australian Government (de Vos, 2021), the current purchasing cost of green hydrogen, including transportation cost (delivered cost per unit), varies between AU$4.9 to AU$8.6 per kg. This cost depends on the amount of hydrogen purchased. The cost of per kg hydrogen (delivered cost per unit) could vary between AU$ 2.6 and AU$ 6 by 2030 due to the reduction of the costs associated with the electrolysis process (de Vos, 2021). In Council and Company (2021) and Gardiner and McClelland (2021), it is predicted that the shipping cost of one kg of green hydrogen in 2030, delivered to the industry users within a short distance (close to 1000 km), will be AU$6.4 (the hydrogen production cost will be AU$3.8). Accordingly, the associated cost of hydrogen shipment is estimated to be AU$ 2.6. These numbers are based on large-scale hydrogen shipment via trucks and pipelines and a significantly longer distance than Horn Island and Thursday Island (approximately 3 km).

Due to the geographical location of the islands, the option of shipping hydrogen by small vessels is more viable. However, estimation of the realistic cost for short distant vessels carrying hydrogen is challenging right now due to the lack of data. Addressing this issue for such a remote location will be a part of future research. Therefore, this study has considered a short-distance subsea hydrogen transmission pipeline to account for the cost of hydrogen transportation for the analysis. The typical cost of building a new subsea hydrogen transmission pipeline varies between US$ 4.7 and US$ 7.1 (approximately AU$ 6.35 and AU$ 9.59) million per km, depending on the distance and location of transmission (Council and Company, 2021). The capital cost of installing a hydrogen transmission pipeline from Horn Island to Thursday Island is considered as AU$19.04 million (due to a very short distance, AU$6.35 million per km is chosen and multiplied by 3 km of the distance between Horn and Thursday Island).

Fig. 6 demonstrates a visual presentation of the overall project. The Thursday Island block presents the electricity generation system, whereas the Horn Island block presents the green hydrogen production system.

3. Methodology

This section describes the steps used in this techno-economic study.

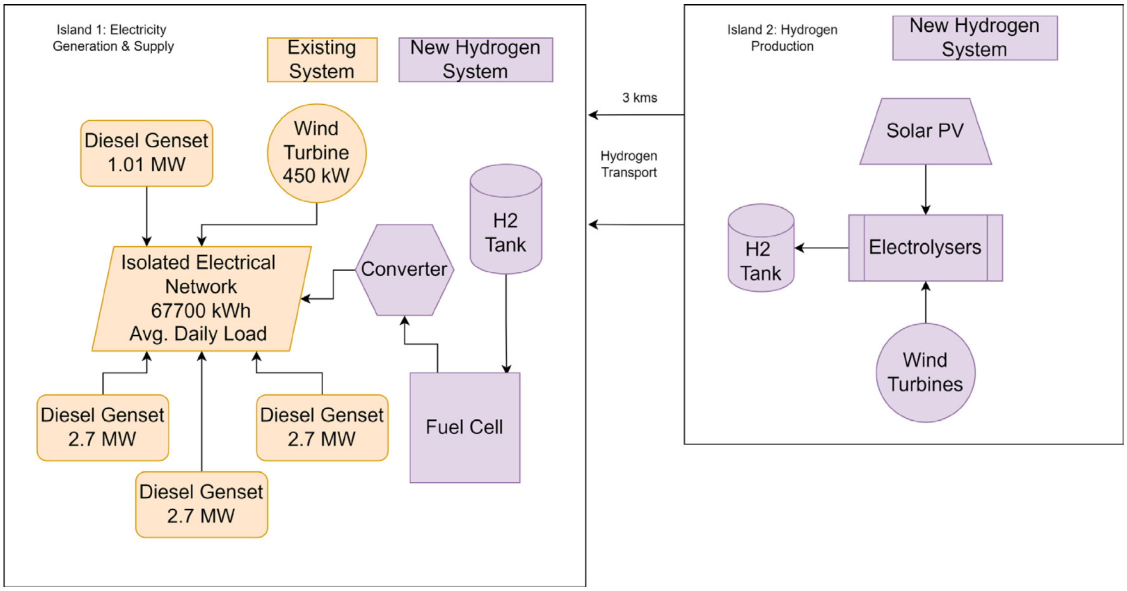

Fig. 6. Overall project illustration.

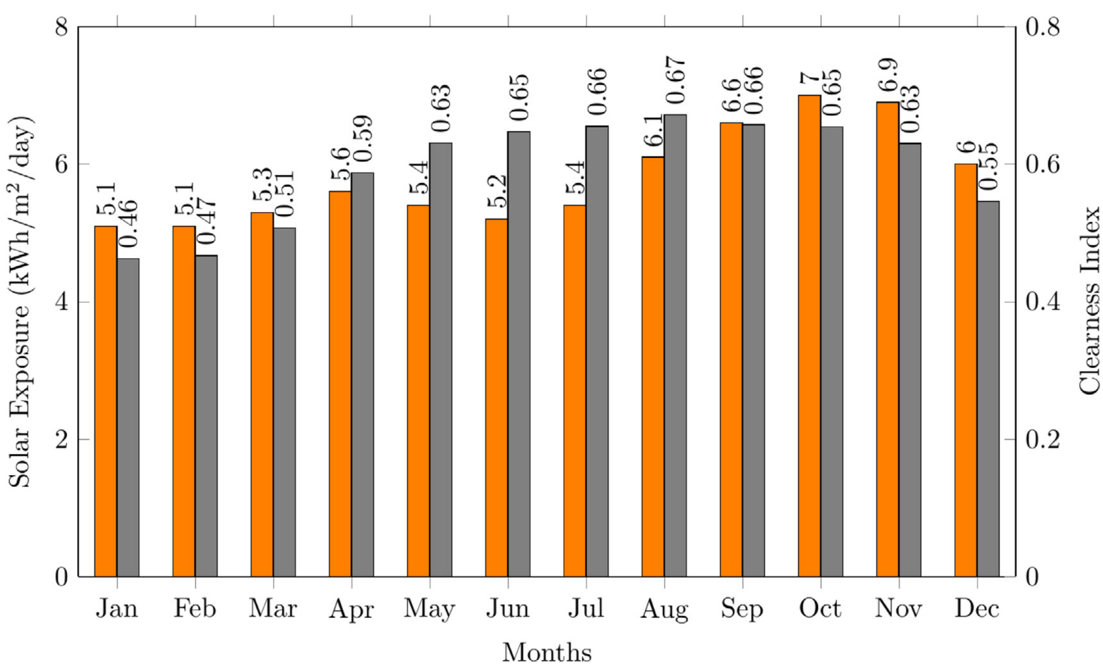

Fig. 7. Average monthly solar exposure on Horn Island.

3.1. Resource data collection

Required data for the study have been obtained from the distribution system operation market search and literature. In addition, the weather data for Horn Island has been collected from the BOM and the NASA. Figs. 7, 8, and 9 present the average monthly solar radiation, wind speed, and temperatures, respectively, at Horn Island. These data are used as the inputs to the model. It is evident from the figures that Horn Island has been exposed to higher solar radiation from August to December, whereas the wind speed is higher during the period of April to October.

3.2. Techno-economic system model

The system models shown in Fig. 6 are implemented in HOMER. To develop the model, weather information, location information, load demand, available resources, and external economic information are used as inputs. The software calculates the optimised size of the system components and the least-cost solution, which also helps to analyse the best scenarios among all available options. The sensitivity of different inputs to outputs is also analysed in this project. Sensitivity analysis is the change in optimisation results in accordance with alterations in one or multiple input parameters. Fig. 10 presents the modelling steps in HOMER.

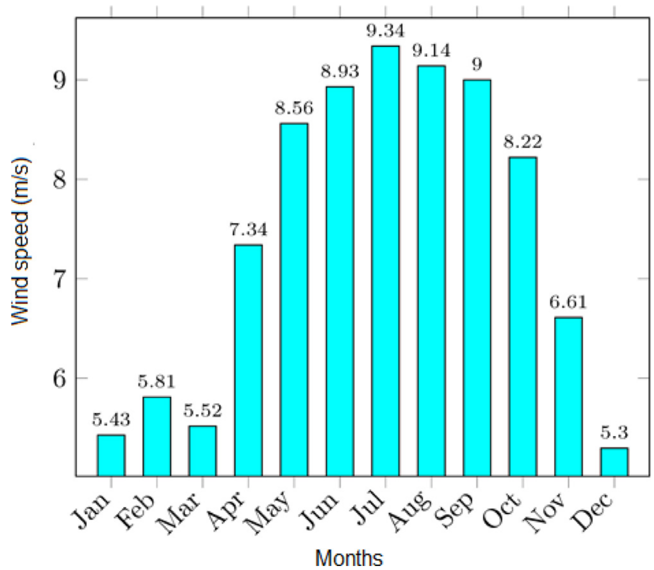

Fig. 8. Average monthly wind speed on Horn Island.

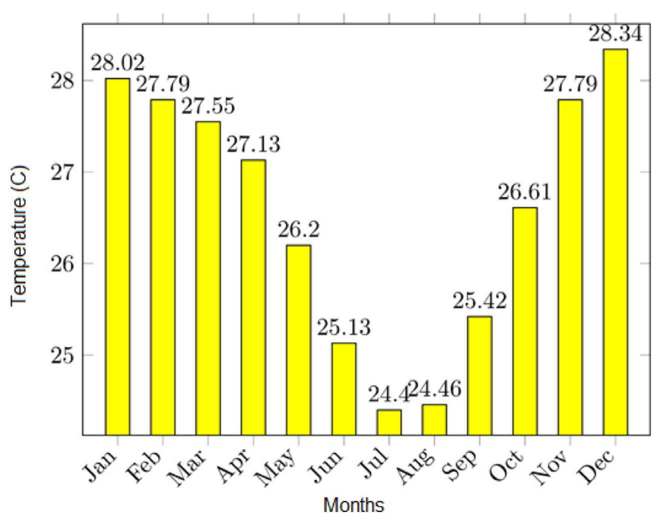

Fig. 9. Average monthly temperature on Horn Island.

In this project, several economic parameters, such as LCOE and NPC, are used to assess different scenarios and identify their economic suitability. These two parameters are two major determinants of the least cost scenario. NPC is the parameter that shows the total cost spent throughout the project lifetime. This is also known as LCC. There are several ways to calculate LCC. However, it can be derived from NPV. An NPV is the sum of the discounted future cash flows, considering both costs and profits (Santos et al., 2019). According to (Osborne, 2010; Santos et al., 2019), NPC can be expressed as,

$$NPV = \frac{NetCashFlow}{(1+f)^t} - I, \tag{1}$$

where *NetCashFlow* is *revenue* − *cost* throughout the project life. NPV becomes NPC when revenue terms are not considered. Therefore in (1), *NetCashFlow* is negative as life cycle cost does not consider any revenue terms. Additionally, f is the interest or discount rate, I is capital investment (initial investment) including price of all components, and t (year) is the time.

LCOE is the average cost of generating one kWh of electricity which accounts for all the associated costs throughout the project lifetime (Rezaei et al., 2021). LCOE can be calculated as,

$$LCOE = \frac{\sum_0^n \frac{I+OM+R}{(1+f)^i}}{\sum_1^n \frac{EWT_1}{(1-d)^{i-1}}}, \tag{2}$$

where OM is the annual operation and maintenance cost, R is the replacement cost of components, n (year) is the project lifespan, d is the degradation factor of components, and EWT_1 is the total amount of electrical energy obtained after one year (in kWh) (Rezaei et al., 2021).

Additionally, the LCOH production is another important factor. The LCOH can be obtained as in (3).

$$LCOH = \frac{C_{elh} + C_{el}}{\sum_1^t M_h}, \tag{3}$$

where t (year) is the lifetime of electrolysers, M_h is the amount of hydrogen produced from electrolysers in kg, C_{el} is the total cost of electricity to run the electrolysis, and C_{elh} is the total cost of electrolysis which can be calculated as,

$$C_{el} = LCOE_h \times \frac{\sum_1^t EWT_i}{t}, \tag{4}$$

$$C_{elh} = C_{u,e} \times \frac{M_h + PE_e}{t \times CF \times \eta_e \times 8760}, \tag{5}$$

where $LCOE_h$ is the levelized cost of the electricity used to produce hydrogen, $C_{u,e}$ is the unit cost of electrolysers, η_e is the electrolyser efficiency, EWT_i is the total amount of electricity obtained after i_{th} year (in kWh), CF is the capacity factor, and PE_e (kW) is the electric power required to run electrolysis.

Due to the area constraints to build renewable energy facilities on Thursday island and the lack of available hydrogen transportation solutions, Horn Island and Thursday Island models were designed separately. However, the findings from both models are integrated to obtain the results. Fig. 11 shows the schematic diagram of the HOMER model for this analysis. The TI model in the figure demonstrates the Thursday Island model for all scenarios, whereas HI model indicates the Horn Island model. There are three HI models with different hydrogen loads for each scenario. The hydrogen load is used to perform simulations and signifies the required hydrogen production for any particular scenario.

3.3. Individual components for the proposed system

Specifications of individual components are described in this subsection.

3.3.1. Solar PV

A Solar PV systems meet a portion of electrolyser electricity demand. Solar PV systems are considered only on Horn Island. The PV panels are connected to the DC bus. The output power of each PV panel is calculated as a direct function of the cell temperature, the incident radiation, and the derating factor (Vera et al., 2020). The derating factor is an energy loss factor due to wire losses, dust, and others that lead to the reduction of output from the expected value. Table 4 presents the specifications of the solar system used in this work.

Table 4
Solar system specifications.

Cost (AU$/kW)	O&M (AU$/kW)	Derating factor (%)	Efficiency (%)	Lifetime (Years)	Orientation
3350	30	90	16	25	2 Axis

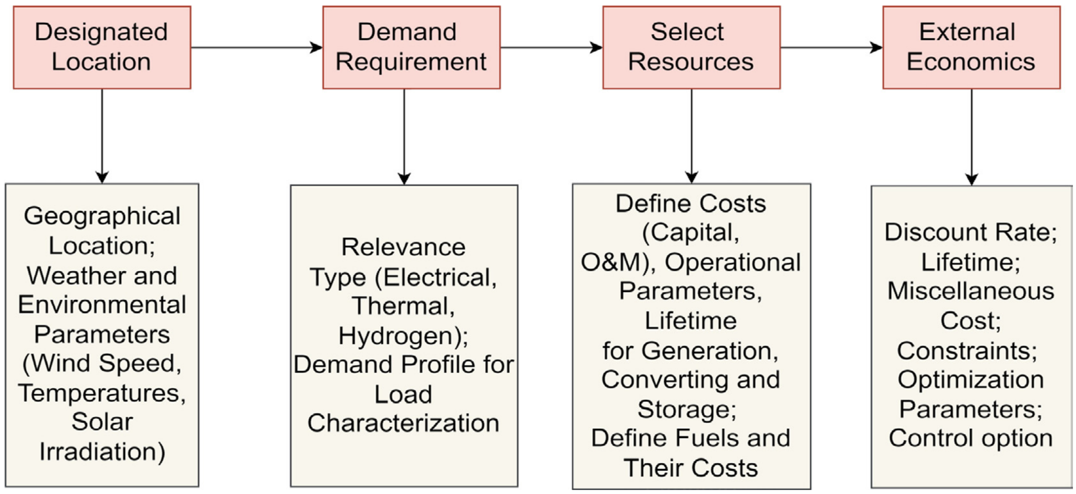

Fig. 10. Modelling process in HOMER.

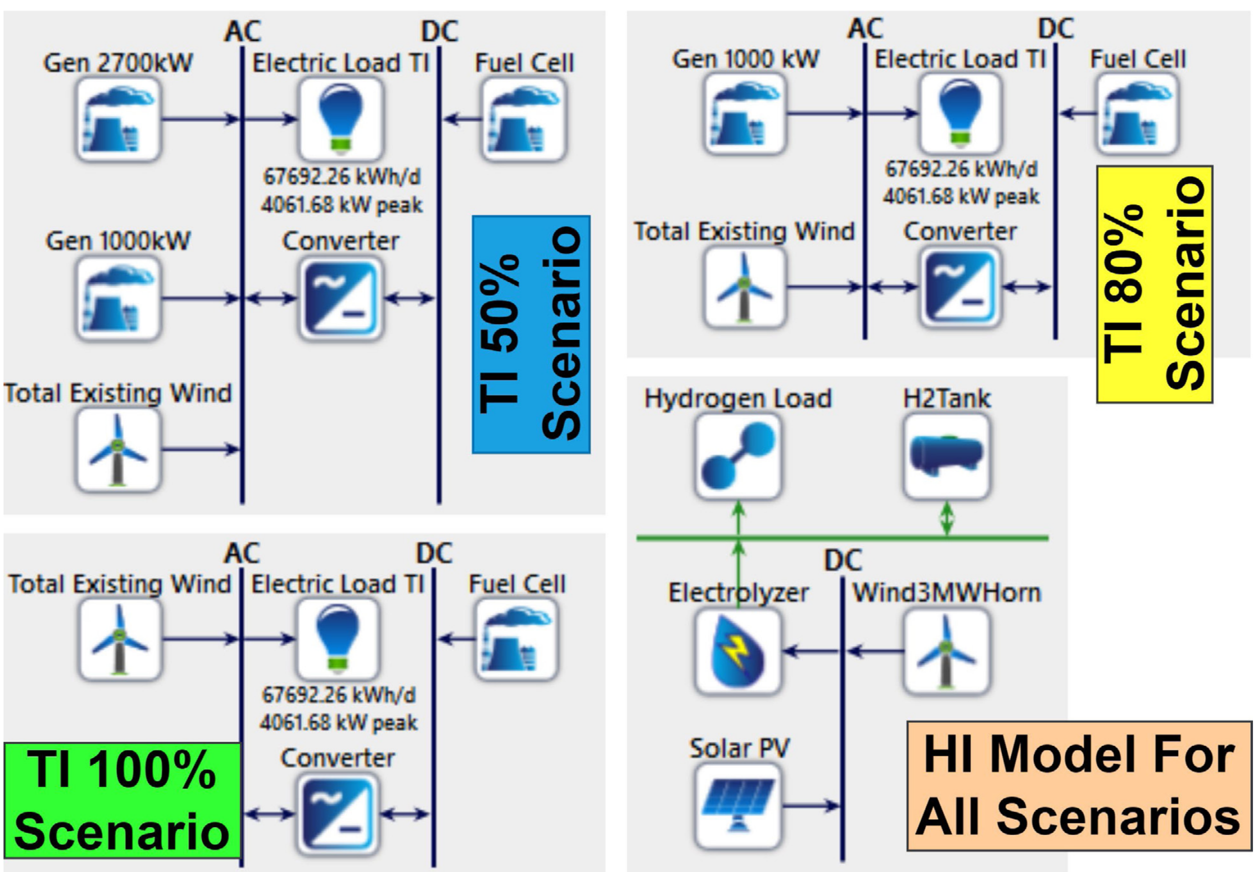

Fig. 11. Thursdays and Horn island models used for simulations in HOMER.

3.3.2. *Wind system*

Wind generators are also considered for the electrolysis as the solar PV systems are insufficient to power Horn Island's entire hydrogen production system. Therefore, a 3 MW Leitwind 101 wind turbine model is selected for this project. Table 5 presents the specifications of the wind generation units used in this study.

Table 5
Wind generation specifications (Ullendullen, 2021).

Cost (AU$/Unit/yr)	O&M (AU$/Unit/yr)	Unit capacity (k W)	Hub height (m)	Lifetime (Years)
7,300,000.00	549	3000	143	20

3.3.3. *Electrolyser*

Electrolyser is a device for performing electrolysis. Electrolysis is a thermochemical process that separates water into hydrogen and oxygen molecules (Ursua et al., 2011). This process contains two electrodes (i.e., anode and cathode) which are separated by a separator. The separator is called a diaphragm (Ursua et al., 2011). The diaphragm is used to prevent the reproduction of water

Table 6
Electrolyser parameters (Navas-Anguita et al., 2020; Schmidt et al., 2017; IRENA, 2020).

Cost (AU$/kW)	O&M (AU$/kW/yr)	Minimum load ratio (%)	Efficiency (%)	Lifetime (Years)
4226.00	0.65	35	80	15

Table 7
Hydrogen tank parameters (Vera et al., 2020).

Cost (AU$/kg)	O&M (AU$/kg/yr)	Initial Tank Level (%)	Lifetime (Years)
1362.00	68	10	25

Table 8
Fuel cell parameters (Mekhilef et al., 2012; U.S. Department Of Energy, 2021).

Cost (AU$/kW)	O&M (AU$/kW/yr)	Efficiency (%)	Lifetime (h)
2000.00	0.01	60	50,000

from separated molecules. Table 6 presents the parameters of the electrolysis used in this study. The electrolyser efficiency could vary between 50% to 83% (Okundamiya, 2021; IRENA, 2020). The efficiency of electrolysers can be obtained as in (6) and (7).

$$\eta\% = \frac{\text{Higher heating value of } H_2}{\text{Energy required per KG of } H_2 \text{ Production}} = \frac{39.4 \text{ kWh/kg}}{\text{Energy required per KG of } H_2 \text{ Production}}. \tag{6}$$

According to Eq. (6), energy required per kg of hydrogen production can be calculated as,

$$\text{Energy required per kg of } H_2 \text{ Production} = \frac{39.4 \text{ kWh/kg}}{\eta\%}. \tag{7}$$

3.3.4. Hydrogen tank

A hydrogen tank is required to store the green hydrogen. The hydrogen tank does not lose efficiency over time and acts similar to conventional gas tanks. Table 7 presents parameters of the hydrogen tank used in this study.

3.3.5. Fuel cell

Hydrogen fuel cells convert the chemical energy of hydrogen into electrical energy. An anode and cathode are separated in a fuel cell by an electrolyte membrane. A catalyst in an anode oxidises supplied hydrogen. Therefore, it produces both electrons and protons. Under the operating condition, protons move towards the cathode of the membrane. The electrons are bypassed towards the cathode via an electrical circuit. The photons form water at the cathode with the oxygen from the air. The passage of electrons via the circuit generates electricity.

There are several types of fuel cells. The cost and efficiency of a fuel cell vary based on its model. The efficiency of a fuel cell can vary between 40% to 85%, whereas the per kW cost of a fuel cell ranges between AU$2000 to AU$4200 (Mekhilef et al., 2012). Although fuel cell efficiency is higher than diesel generators, the per kW cost of the fuel cells is five times more expensive than diesel generators (Mekhilef et al., 2012). Moreover, a fuel cell has a lower lifespan. Depending on the operating condition, the lifespan of a fuel cell varies from 5 to 20 years (Sharaf and Orhan, 2014). The lifespan of a fuel cell should reach approximately 80,000 h to be economically viable (Sharaf and Orhan, 2014). Table 8 presents the parameters of the fuel cell used in this study.

Table 9
Converter cell parameters (Battelle, 2017; Koo et al., 2021).

Cost (AU$/kW)	O&M (AU$/kW/yr)	Efficiency (%)	Lifetime (Years)
300.00	25.00	95	15

Table 10
Assumptions for the analysis.

Items/Parameters	Assumed value
Diesel generators	Cost: Zero (as already exist) Capacities: 1.01 MW & 2.7 MW
Wind generators	Cost: Zero (as already exist)
Diesel price (terminal gate)	AU$1.3 per litre
Energy density of 1 kg of hydrogen	33.36 kWh (Handwerker et al., 2021)
Unmet hydrogen production	0%
Discount rate	4.07%
Inflation rate	1.5%
Project lifetime	25 years
Additional costs (costs not associated with the components)	Not considered

Table 11
Daily average electricity required for each scenario.

Scenarios	Required renewable electricity (Daily)
100% renewable	60,930 kWh
80% renewable	48,744 kWh
50% renewable	30,465 kWh

3.3.6. Converter

An electrical power system with a fuel cell requires converters to connect to the distribution network (since fuel cells provide DC as the output power). A converter converts the DC to AC. Such a converter is also called an inverter. The fundamental operation characteristics of the inverter can be expressed as in (8) (Gopinath et al., 2004).

$$V_{AO} = m_a \times V_{DC} 2\sin(\omega_1 t), 0 < m_a < 1 \tag{8}$$

where V_{AO} (V) is the inverter output voltage, V_{DC} (V) is the fuel cell output voltage, and $0 < m_a < 1$ is the modulation index. The inverters used for fuel cell power system operation are highly efficient (close to 95% efficiency) (Koo et al., 2021; Vera et al., 2020). A generic HOMER converter has been used for this research work. Table 9 presents the parameters of the converter system used in this paper.

4. Results and discussions

This section presents the results of the numerical analysis. Some assumptions have been made for this numerical analysis which are given in Table 10. As discussed in Section 2, the daily average electrical load demand on Thursday Island is 67,692 kWh. The 10% demand is assumed to be met by the two existing wind generators. Thus, the remaining 60,930 kWh demand must be generated by other energy sources.

Three renewable scenarios have been selected for this work. Table 11 presents the scenarios and the amount of daily average electricity demand that has to be generated by RES for each scenario.

4.1. Base model on Thursday Island

The base model is designed to analyse the existing electricity generation system of Thursday Island. It is assumed that the remaining life of diesel gensets is 40,000 h. Since the diesel gensets and the wind turbines are already commissioned and in operation on Thursday Island, the CAPEX of those generator components

Table 12
Techno-economic summary of existing generation units.

Yearly cost of operating wind system	Cost of operating diesel genset	Farm gate diesel price per litre	LCOE
AU$183.00	AU$262	AU$1.3	AU$0.512

Table 13
Optimised capacity (100% renewable scenario).

Components/Parameters	Capacity	Unit
Electrical load	60 930	kWh
Required hydrogen production	3044.1	kg
Solar PV	30	MW
Wind generation	33	MW
Electrolyser	15	MW
Hydrogen tank	4000	kg
NPC_h (AUD)	431	Million
Capex (AUD)	250	Million

Table 14
Optimised capacity (80% renewable scenario).

Components/Parameters	Capacity	Unit
Electrical load	48 744	kWh
Required hydrogen production	2435.3	kg
Solar PV	20	MW
Wind generation	21	MW
Electrolyser	12	MW
Hydrogen tank	4500	kg
NPC_h (AUD)	299	Million
Capex (AUD)	175	Million

is considered zero. The average cost of hourly operation of each genset and wind generator is AU$ 262 and AU$ 183, respectively. That includes all associated costs. Wind speed data for Thursday Island has been used for this base model simulation. Table 12 provides the economic summary of the existing generation units. The LCOE is estimated as AU$ 0.512/kWh based on the given information.

4.2. Hydrogen production

The amount of produced hydrogen for each scenario has been calculated and simulated as hydrogen loads. The hydrogen load has been distributed hourly, depending on the daily hydrogen requirements of Thursday Island fuel cells. Tables 13, 14, and 15 present the optimal renewable capacity of each scenario. These tables also illustrate the capacity of the hydrogen production system and overall total cost, where NPC_h is the net present cost of the Horn Island model. For example, to achieve the 100% renewable scenario, 30 MW of solar PV and 11 units of 3 MW wind generators are required. For the 80% scenario, 20 MW of solar PV and seven units of 3 MW wind generators are necessary, whereas, for the 50% scenario, 10 MW of solar PV and five units of 3 MW wind generators should be used. Fig. 12 presents a visual illustration of the optimal capacity of the components for the hydrogen production system used in all scenarios. These scenarios are illustrated in different colours. Fig. 12 summarises the values given in Tables 13 to 15.

It is worth mentioning that the presented simulation results were obtained based on the context of producing 100% required hydrogen and considering 0% unmet load regardless of the resource scarcity due to weather condition. This might result in oversizing the optimal capacity of the solar PVs and the wind generation. If existing diesel generators are used as reserved generation capacity due to the shortage of hydrogen production caused by weather condition, the overall capacity of RES can be reduced as well as the excess electricity and its related cost. Further analysis of unmet load (unmet hydrogen production) has been studied in sensitivity analysis section.

Table 15
Optimised capacity (50% renewable scenario).

Components/Parameters	Capacity	Unit
Electrical load	30 465	kWh
Required hydrogen production	1522	kg
Solar PV	10	MW
Wind generation	15	MW
Electrolyser	8	MW
Hydrogen tank	3000	kg
NPC_h (AUD)	192	Million
Capex (AUD)	108	Million

4.3. Electricity generation

As stated earlier, 90% of the load on Thursday Island is currently supplied by diesel generators. Therefore, any alteration in sources of electricity generation causes a change in component size and LCOE. Table 16 represents the required fuel cell capacity for all scenarios considered in this paper. Depending on renewable scenarios, fuel cell capacity could vary between 2.5 and 4.5 MW. For modelling and further simplicity, the smallest size of the existing genset (i.e., 1.01 MW) is considered for an 80% renewable scenario, whereas an additional 2.7 MW existing genset is also considered for a 50% scenario.

Table 16 also shows the LCOE and NPC of a hydrogen-based electricity generation system on Thursday Island. The current LCOE of the Thursday Island system is AU$ 0.512/kWh with the four diesel systems of 2.7 MW each. However, stepping towards a greener system provides LCOE of AU$ 1.08/kWh, AU$ 0.78/kWh, and AU$ 0.59/kWh, respectively, depending on the percentage of green hydrogen penetration. These numbers can vary depending on the cost of hydrogen transportation and diesel price.

Table 17 compares CO_2 emissions and hydrogen consumption by the electricity generation system on Thursday Island. The environmental cost of the entire system reduces significantly with the inclusion of the hydrogen system. For instance, in the 100% renewable scenario, 17,607,077 kg of CO_2 emissions per year could be possible, whereas 50% and 80% scenarios can reduce yearly emissions by 78% and 97%, respectively.

4.4. Sensitivity analysis

A sensitivity analysis oversees the change in output parameters due to any alteration in a particular input parameter. In this project, six parameters have been selected for the sensitivity study. These are unmet hydrogen production, hydrogen transportation cost, diesel cost, PV derating factor, PV without the tracking system, electrolyser efficiency, and undersea cable. Table 18 presents the factors considered for the sensitivity analysis in this work. The table also shows the renewable penetration scenarios considered for each sensitivity analysis where 'All' means the sensitivity analysis has considered all renewable penetration scenarios, i.e., 100%, 80%, and 50% renewable.

4.4.1. Impact of unmet hydrogen production

The unmet hydrogen production means less hydrogen is produced from the Horn Island system than the required hydrogen for fuel cells on Thursday Island in case of unlikely weather issues for the 100% renewable scenario. A sensitivity analysis has been performed considering 0% to 20% unmet hydrogen production. The changes in capacity and costs are compared in Table 19. In summary, a backup facility to support the system (in case of producing up to 20% less hydrogen due to RES unavailability) would reduce the solar and wind capacity by approximately 67% and 73%, respectively. Therefore, the Horn Island model's life cycle cost or net present cost (NPC) and the overall LCOE would

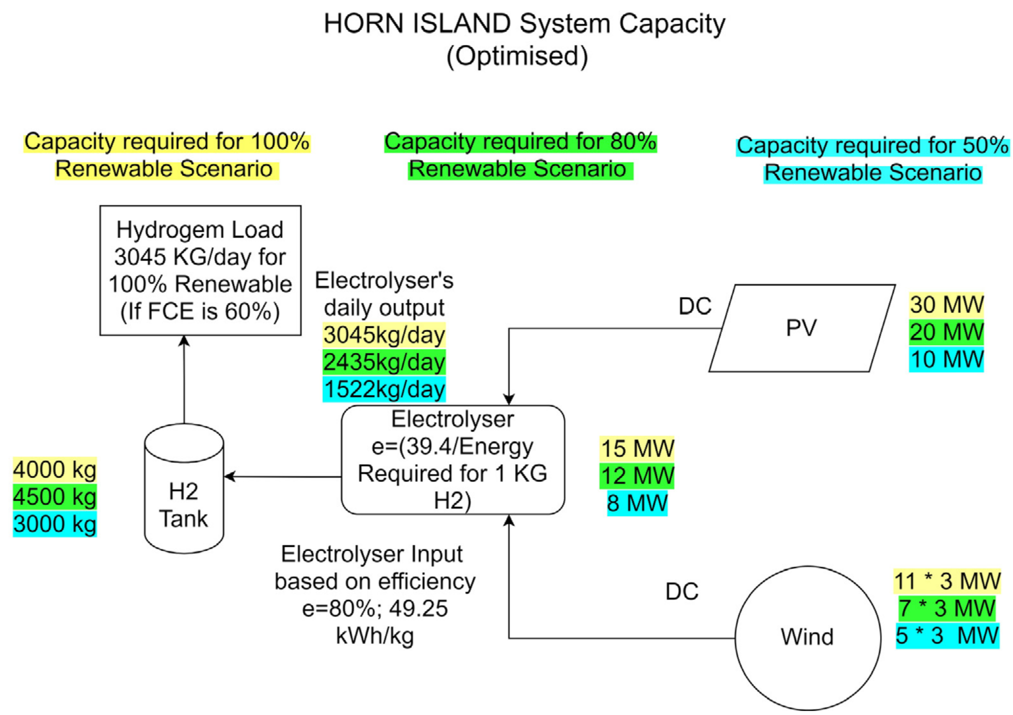

Fig. 12. Optimised component capacity of the green hydrogen production system on Horn Island.

Table 16
Capacity of fuel cell versus diesel generation in each scenario.

Renewable scenarios	Fuel cell capacity (MW)	Converter capacity (MW)	Diesel genset (MW)	LCOE (AU$)	NPC (AU$)
100% renewable	4.5 MW	3.94 MW	0	1.08	122M
80% renewable	3.6 MW	3.43 MW	1.01 MW	0.78	117M
50% renewable	2.5 MW	2.54 MW	1.01 MW & 2.7 MW	0.59	125M

Table 17
Yearly CO_2 emission for each scenario.

Renewable scenarios	Hydrogen consumption (kg/year)	CO_2 emission (kg/year)	CO_2 reduction (in %)
100% Renewable	1,270,607	Almost 0	Almost 100%
80% Renewable	1,242,186	529,228	96.99%
50% Renewable	1,047,595	3,913,357	77.77%
0% Hydrogen	0	17,607,077	N/A

reduce by 61.7% and 54.2%, respectively. It is worth mentioning that the LCOE of the 100% renewable scenario, considering 20% unmet hydrogen production, is AU$ 0.50/kWh, which is less than the LCOE of the existing electricity system (compared to the base model, Table 12). Table 19 demonstrates these changes based on a 100% renewable scenario.

Figs. 13, 14, and 15 present time series curves of 0%, 10%, and 20% unmet hydrogen production, respectively. In these figures, the black line presents hydrogen demand to be met, the green line is the electrolyser output, and the red line represents the hydrogen storage. Figs. 14 and 15 present the declining trends of hydrogen storage when unmet hydrogen is 10% and 20%, respectively. It can be seen that the storage can drop up to zero depending on the weather condition. However, as seen in Fig. 13, a higher storage level is required to maintain the 0% unmet hydrogen, which results in an extensive hydrogen production system on Horn Island due to the renewable intermittency. Therefore, the capacity of solar PV and wind generators becomes higher to meet the demand under variable weather conditions. That concludes in higher excess electricity as well as a higher

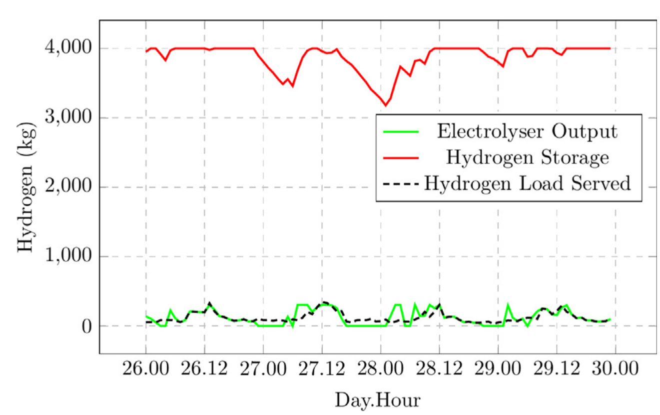

Fig. 13. Time series curves for 0% unmet hydrogen production in April. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

level of hydrogen storage in the Horn Island system. Suppose the existing diesel generators are kept as the backup generator for up to 20% of unmet hydrogen production. In that case, the overall component sizing of the hydrogen production system can be reduced significantly. Therefore, the overall cost of the system is reduced.

Though this section mainly focuses on a 100% renewable scenario, it is worth providing a summary of the impact of unmet hydrogen production for 80% and 50% renewable scenarios. Table 20 presents the changes in sizes of solar PV and wind generators due to unmet hydrogen under 80% and 50% renewable

Table 18
Factors considered for sensitivity analysis.

Sensitivity	Range of input factors	Output factors	Scenarios
Unmet hydrogen production	0% to 20%	NPC, LCOH, LCOE & Component size	100% Renewable
Hydrogen transportation maintenance cost	0% to 10% of capital/year	LCOE	All
Diesel cost per litre	AU$0.9 to AU$1.3	LCOE & Capacity	80% & 50% Renewable
PV derating factor	75% to 95%	NPC & LCOH	All
PV tracking system	Two-axis tracking off	Capacity, NPC & Capex	All
Electrolyser efficiency	70% to 90%	LCOH	All
Undersea cable consideration	Undersea Cable vs. Pipeline	LCOE	All

Table 19
Changes in costs and capacities due to the unmet load in 100% renewable scenario.

Unmet H_2 load	PV capacity	Wind capacity	Excess electricity	LCOH (AU$)	NPC_h (Million AU$)	LCOE (AU$)
0%	30 MW	33 MW	72%	20.4	431M	1.08
10%	18 MW	9 MW	35.6%	11	209M	0.59
20%	10 MW	9 MW	23.7%	9.72	165M	0.50

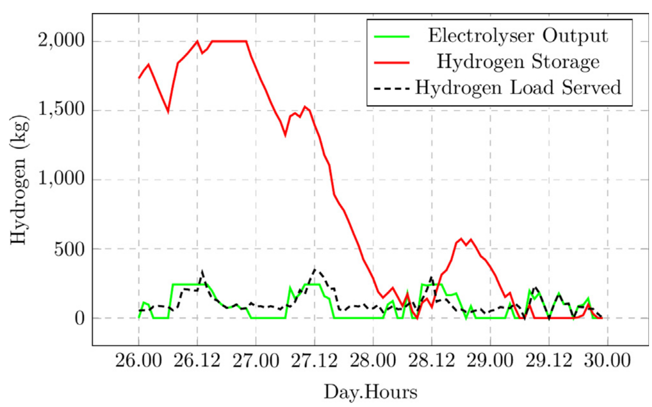

Fig. 14. Time series curves for 10% unmet hydrogen production in April. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

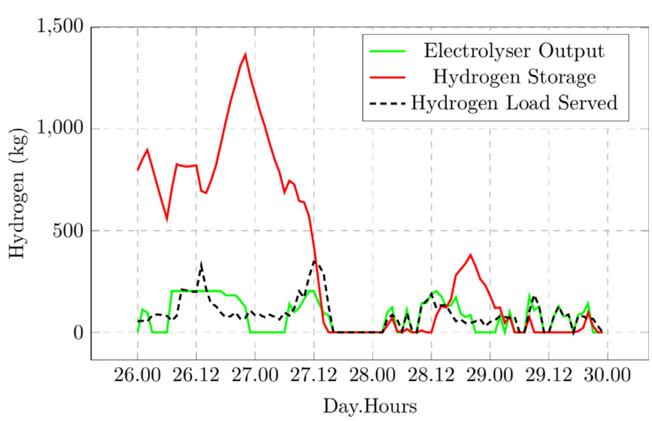

Fig. 15. Time series curves for 20% unmet hydrogen production in April. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

scenarios. The table also shows the changes in LCOE, LCOH, and energy loss. For 10% unmet production, the overall size of RES reduces by 58% in the case of the 80% renewable scenario. The total size of RES reduces by 50% in the case of the 50% renewable scenario (compared to the results given in Tables 14 and 15). Similarly, for 20% unmet hydrogen production, the overall size of RES reduces by 68% (80% renewable scenario) and 60% (50% renewable scenario), respectively (see Tables 14 and 15). Reduction in generation component sizing reduces the overall costs of the system. The LCOE varies between AU$ 0.37/kWh and AU$ 0.48/kWh for unmet hydrogen of 10% and 20%. The LCOE range suggests a lower cost of electricity generation with a green hydrogen system than the cost of electricity generation from the prevailing system (compared to the base model, Table 12).

Table 20
Change in size and cost of the system due to unmet hydrogen consideration for 80% and 50% renewable scenarios.

Scenario	80% Renewable		50% Renewable	
Unmet hydrogen production	10%	20%	10%	20%
Solar PV capacity	12 MW	5 MW	3.5 MW	10 MW
Wind generation capacity	9 MW	9 MW	9 MW	0
Excess electricity (Energy loss)	38.6%	25%	43.6%	9.32%
LCOH (AU$)	10.9	9.97	11.5	11.1
LCOE (AU$)	0.48	0.41	0.40	0.37

4.4.2. Sensitivity of transportation cost

A subsea pipeline has been considered to transfer hydrogen from Horn Island to Thursday Island. The capital cost of installing the pipeline is already incorporated in the simulation. However, the yearly cost of operation and maintenance of the pipeline has yet to be factored into the calculations. This section presents a sensitivity analysis of the pipeline's operation and maintenance (O&M) cost. The yearly O&M cost is assumed to vary from 0% to 10% of the total capital cost of the subsea pipeline. Fig. 16 presents the change in LCOE due to the changes in the transportation cost (yearly O&M cost) of hydrogen. The figure shows upward trends of LCOEs for the rise in hydrogen transportation cost from 0% to 10%. However, the figure also demonstrates that the LCOE is less sensitive to the yearly O&M cost of hydrogen transport for 80% and 50% renewable scenarios than the 100% scenario.

4.4.3. Diesel price sensitivity

This study has considered the terminal gate price of diesel. However, numerous global and domestic factors can cause a change in diesel prices throughout the project progression. Therefore, performing a sensitivity analysis of diesel price fluctuation is beneficial. Table 21 shows the impact of diesel price change on component sizing and LCOE. Three flat prices of diesel have been selected from AU$ 0.90 to AU$ 1.1 for this sensitivity. This price could be further reduced for industries with bulk requirements. As can be seen, the change in diesel price has no or very insignificant impact on component sizing and LCOE in the case of the 80% scenario. However, this impact is more evident in the case of the 50% renewable scenario since this scenario has a greater dependency on diesel. Moreover, LCOE reduces by the reduction of diesel prices.

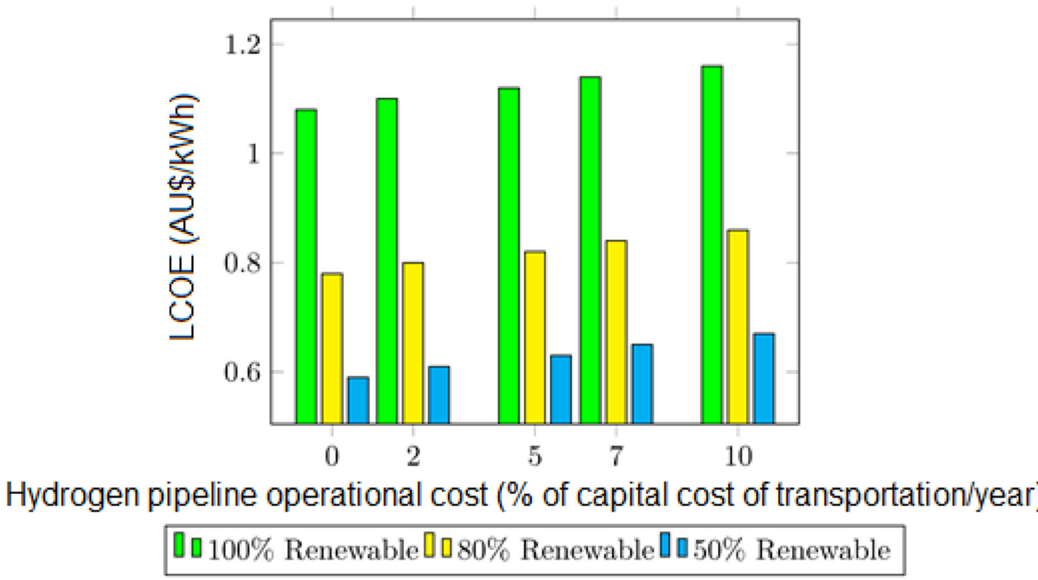

Fig. 16. Impact of change in transportation cost on LCOE.

Table 21
Cost and capacity change of the system due to change in diesel price.

Diesel price	80% Renewable			50% Renewable		
(AU$/l)	Fuel cell	Converter	LCOE	Fuel cell	Converter	LCOE
0.9	3.6 MW	3.43 MW	0.780	2.5 MW	2.38 MW	0.569
1	3.6 MW	3.43 MW	0.781	2.5 MW	2.38 MW	0.575
1.1	3.6 MW	3.43 MW	0.782	2.5 MW	2.38 MW	0.581
1.3	3.6 MW	3.43 MW	0.782	2.5 MW	2.54 MW	0.593

4.4.4. Impact of various PV derating factors

Fig. 17 shows the impact of the solar derating factor on NPC of the Horn Island system, whereas Fig. 18 illustrates the impact of the solar derating factor on LCOH. Five derating factors varying from 75% to 90% were selected in the simulation. The solar PV derating factor is the reduction of output power from the rated value over the years (HOMER, 2021), also known as the solar degradation factor. The derating factor of a solar PV panel also considers the impact of outdoor weather conditions on the panel (Pan et al., 2011). Figs. 17 and 18 show the changes in NPC and LCOH for three renewable scenarios. For the 80% renewable scenario, the minimum and lower quartile, as well as the maximum and higher quartile, are the same. Therefore, under the 80% renewable scenario, the impact of 70% and 75% derating factors on the costs is the same. Similarly, the impact of 85% and 90% derating factors is also the same. In the 50% renewable scenario, the impact of 75%, 80%, and 85% derating factors on the costs remains the same.

4.4.5. Impact of non-tracking solar PV system

Higher cost of maintenance, lack of local skilled labour, and a harsh environment affect the usage of two-axis tracking solar PV systems used in the Horn Island system. Therefore, the size of RES shown in Tables 13∼15 might differ. Consequently, it is important to present changes in the size of RES as well as the overall costs when single-axis (no tracking system) solar PVs are used. Table 22 demonstrates these changes. Compared to Tables 13∼15, no difference is noticed under the 50% renewable scenario. However, the total size of RES increases by 2 MW in the 100% scenario, whereas in the 80% scenario, a reduction of 2 MW is observed.

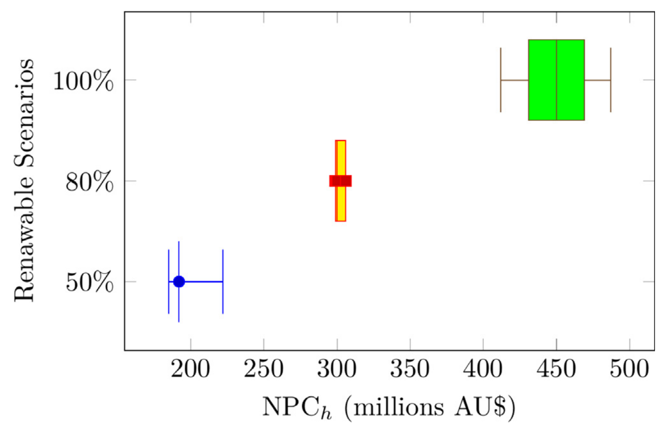

Fig. 17. Impact of PV derating factor on NPC$_h$.

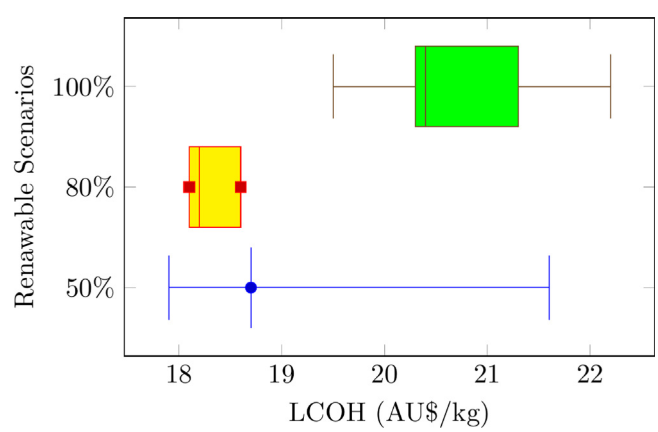

Fig. 18. Impact of PV derating factor on LCOH.

4.4.6. Sensitivity of electrolyser efficiency

Figs. 19, 20, and 21 illustrate the impact of electrolyser efficiency on the cost of producing hydrogen under 0%, 10%, and 20% unmet hydrogen scenarios. The efficiency varied between 70% to 90% for this analysis. From the results in Figs. 19 to 21,

Table 22
Effect of non-tracking system on the capacity of RES (Horn Island).

Scenario	Generation mix		Economics	
	Solar PV	Wind generation	NPC_h (AUD)	Capex (AUD)
100% Scenario	35 MW	30 MW	432 million	259 million
80% Scenario	12 MW	27 MW	308 million	164 million
50% Scenario	10 MW	15 MW	192 million	108 million

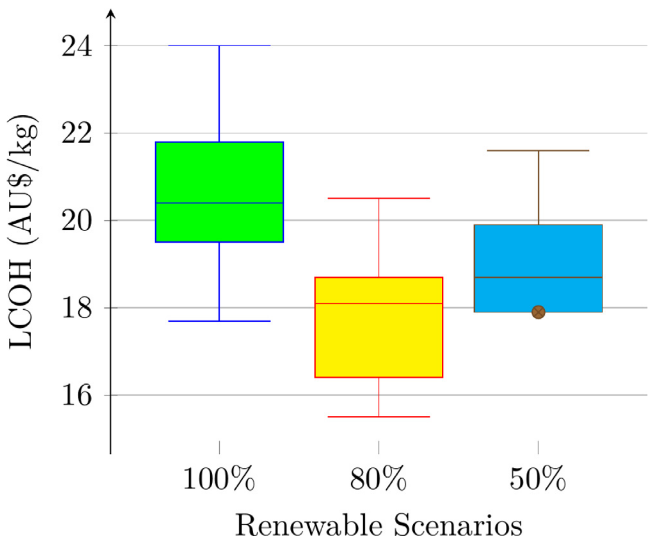

Fig. 19. Impact of electrolyser efficiency variation on LCOH in case of 0% unmet hydrogen production.

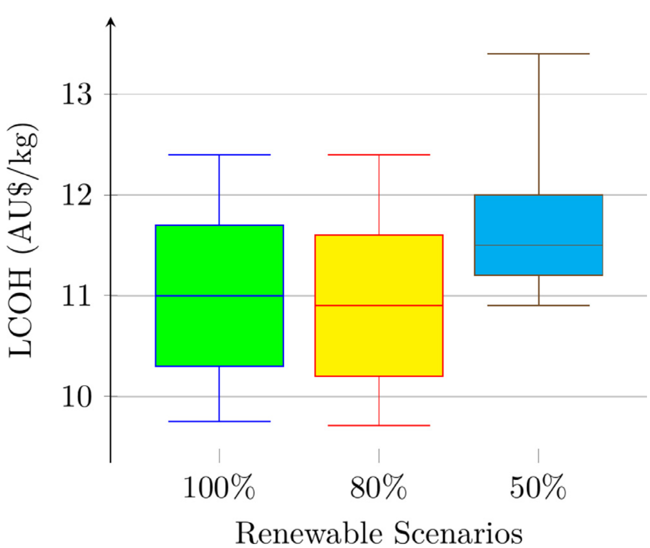

Fig. 20. Impact of electrolyser efficiency variation on LCOH in case of 10% unmet hydrogen production.

it is evident that hydrogen production costs are lower for the 80% renewable scenario. The LCOH has been used as the index for this sensitivity study. In the case of 0% unmet hydrogen production (see Fig. 19), LCOH varies between AU$ 15.5/kg to AU$ 24/kg, whereas, for 10% unmet hydrogen, the LCOH varies between AU$ 9.5/kg to AU$ 13.5/kg depending on the scenario (see Fig. 20). Furthermore, the LCOH varies between AU$ 8.5/kg and AU$ 12.2/kg for the 20% unmet hydrogen scenario.

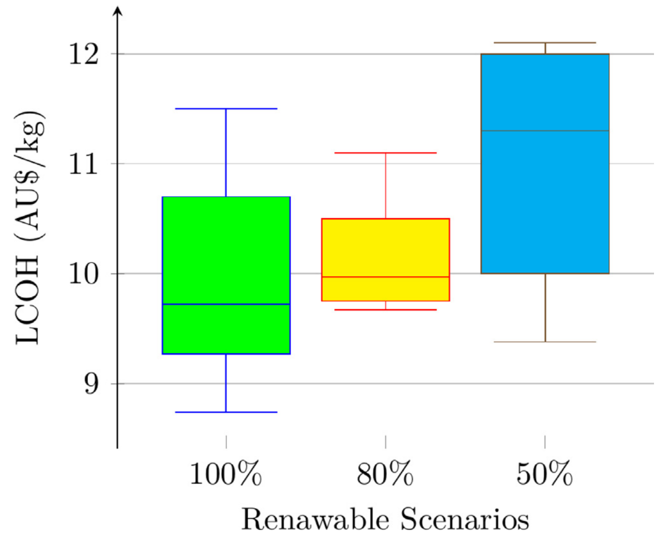

Fig. 21. Impact of electrolyser efficiency variation on LCOH (20% unmet hydrogen production).

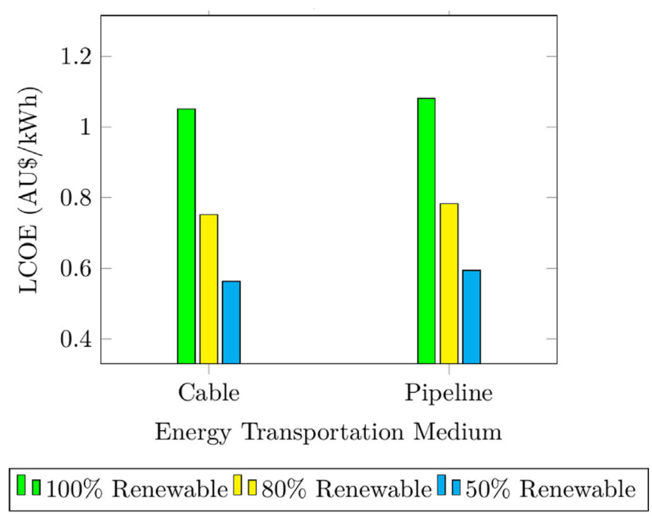

Fig. 22. Impact of change in undersea cable cost on LCOE.

4.4.7. Impact of undersea cable

It is worth calculating the change in cost parameters if an undersea cable is used. This study is performed with an undersea cable instead of considering the hydrogen supply chain (hydrogen pipeline). Fig. 22 presents a cost comparison between the hydrogen pipeline and the undersea cable. For a 100% scenario, an undersea cable can reduce the LCOE by AU$ 0.03/kWh. The LCOE reduction will be AU$ 0.04/kWh and AU$ 0.05/kWh for the 80% scenario and 50% scenario, respectively. However, connecting both islands by undersea cable would require some environmental approval and policy changes.

4.5. Comparative assessment

This section presents the comparative assessment of the proposed model of the microgrid to the microgrid model presented in the literature (Vera et al., 2020; Clairand et al., 2019) for powering islanded power systems. The 80% renewable scenario is considered to accommodate the sources listed (Vera et al., 2020; Clairand et al., 2019). The LCOE has been used as the indices for this analysis. Table 23 shows that the proposed model exhibits the lowest LCOE under the 80% renewable scenario.

Table 23
Comparative analysis results.

Model	LCOE (AU$/kWh)
Proposed model	0.78
Vera et al. (2020);	0.83
Clairand et al. (2019)	0.80

5. Conclusion

This paper has demonstrated a primary techno-economic model of a green hydrogen-based microgrid on Thursday Island, Australia, as part of a Hydrogen Hub project (Hydrogen hub powering remote communities). The major obstacle to design such a model is finding an appropriate space for a green hydrogen production system on the island. The proposed model in this paper can overcome this challenge by placing the hydrogen production system on the neighbouring island, i.e., Horn Island. The findings of this paper elucidate the required capacity of RES for the green hydrogen production system. The simulation results show that out of the three scenarios considered in this paper, the 80% scenario provides a better solution in terms of cost-effectiveness and reduction of CO_2 emissions. In this scenario, the LCOE varies between AU$0.41/kWh to AU$0.48/kWh, which is less than the current generation costs on Thursday Island. The simulation results also demonstrate the lowest LCOH for a 100% renewable scenario when 20% unmet production is considered. Furthermore, a 50% renewable scenario with 20% unmet hydrogen production offers the least amount of excess electricity. This research study provides a techno-economical guideline for hydrogen-based microgrids in remote locations in Australia and the global context where green hydrogen production is not possible at the location of microgrids. An extensive analysis regarding green hydrogen transportation needs to be considered. This work did not consider the uncertainties associated with the costs of components. Furthermore, the policies and government subsidies significantly impact the LCOE and LCOH, which were not assessed in this work. Additional scenarios, such as the financial modelling of importing hydrogen and producing locally, should have been considered in this work. These studies will be conducted in the future and reported accordingly.

CRediT authorship contribution statement

Tanvir Hasan: Conceptualization, Data collection, Software, validation, Writing, editing. **Kianoush Emami:** Reviewing and editing. **Rakibuzzaman Shah:** Conceptualization, survey, Writing, reviewing, and editing. **N.M.S. Hassan:** Reviewing and supervision. **Vitali Belokoskov:** Reviewing, survey and validation. **Max Ly:** Editing and reviewing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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